



Debug Using Serial.print()

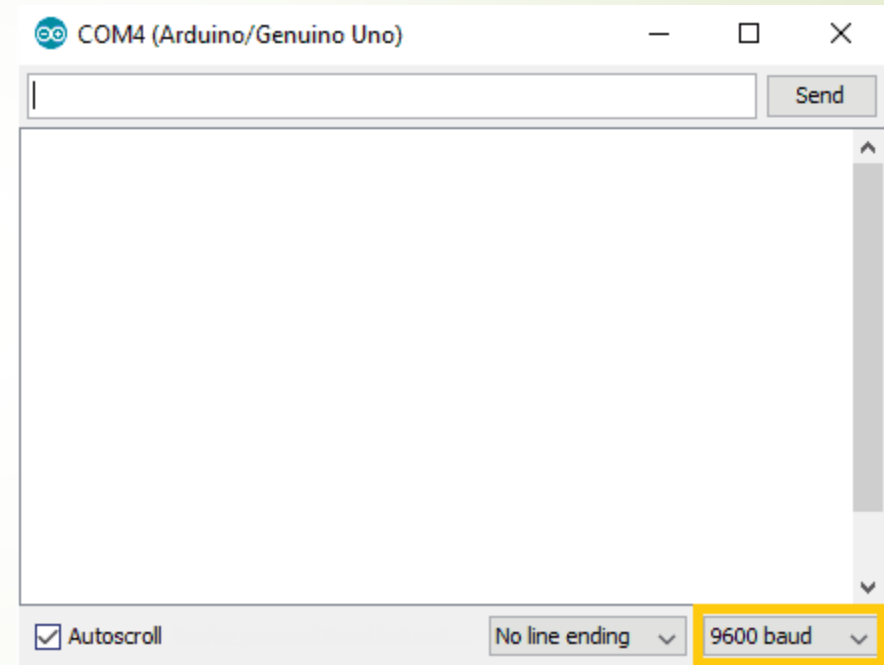
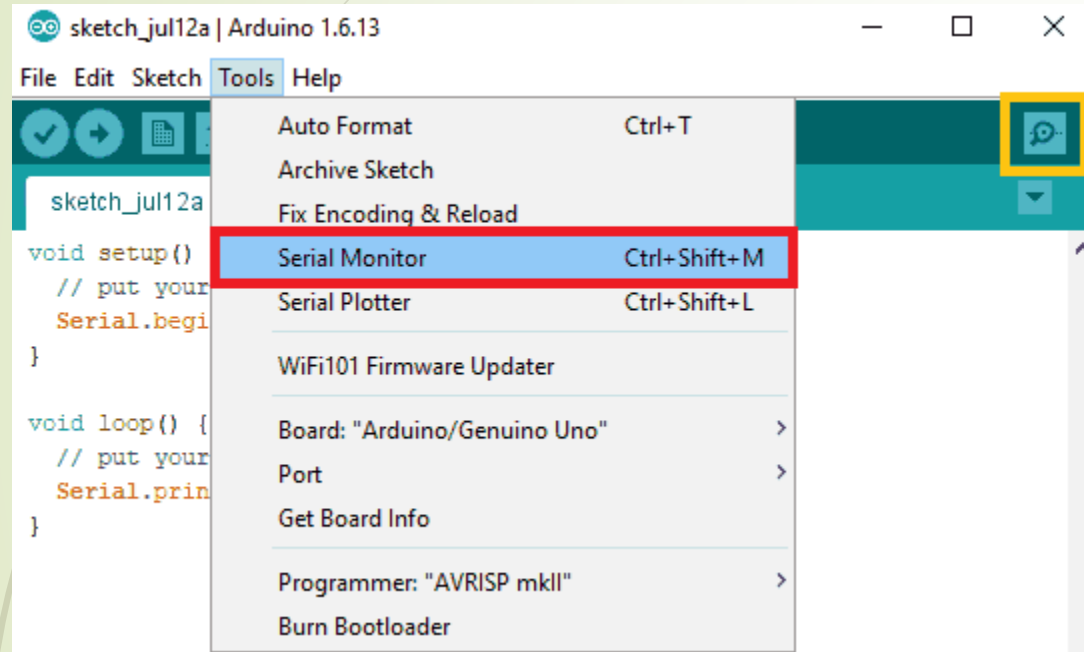
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<https://www.arduino.cc/reference/en/language/functions/communication/serial/print/>



Arduino Serial Monitor

- Arduino IDE has integrated serial monitor which can be used to watch serial data.
- After uploading program to the Arduino, open **Serial Monitor** to watch Serial data.
- You can find Serial monitor



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- One can change baud rate
 - If baud rate of two devices mismatched, then we will get nothing or garbage value.



Functions Used for Serial Communication

- **Serial.begin (baud rate)**

- This function is used to initiate the serial communication with baud rate.
e.g. `Serial.begin(9600)`

- **Serial.print (data, format type (optional))**

- It sends **ASCII character** on serial port so human can read it easily.
- This function converts the data to ASCII character and then send (print) it.
e.g. `Serial.print (123);`
`Serial.print ("123");`
`Serial.print ("ABC");`
`Serial.print (12, HEX)`

- **Serial.println (data, format type (optional))**

- It sends ASCII character on serial port followed by carriage return.

- **Serial.write()**

- This function writes **binary data** to the serial port.

- It returns number of byte written.

- e.g.** Serial.write (65); // it writes actual byte to serial port

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- `Serial.write(0x45);` // will write 0100 0101 to the cable, // **E**
 - `Serial.print(76, BIN)` gives "0100 1100"
 - `Serial.print(76, OCT)` gives "114"
 - `Serial.print("L", DEC)` gives "76"
 - `Serial.print(76, HEX)` gives "4C"

➤ **Serial.available ()**

- This function returns the number of bytes available to read.

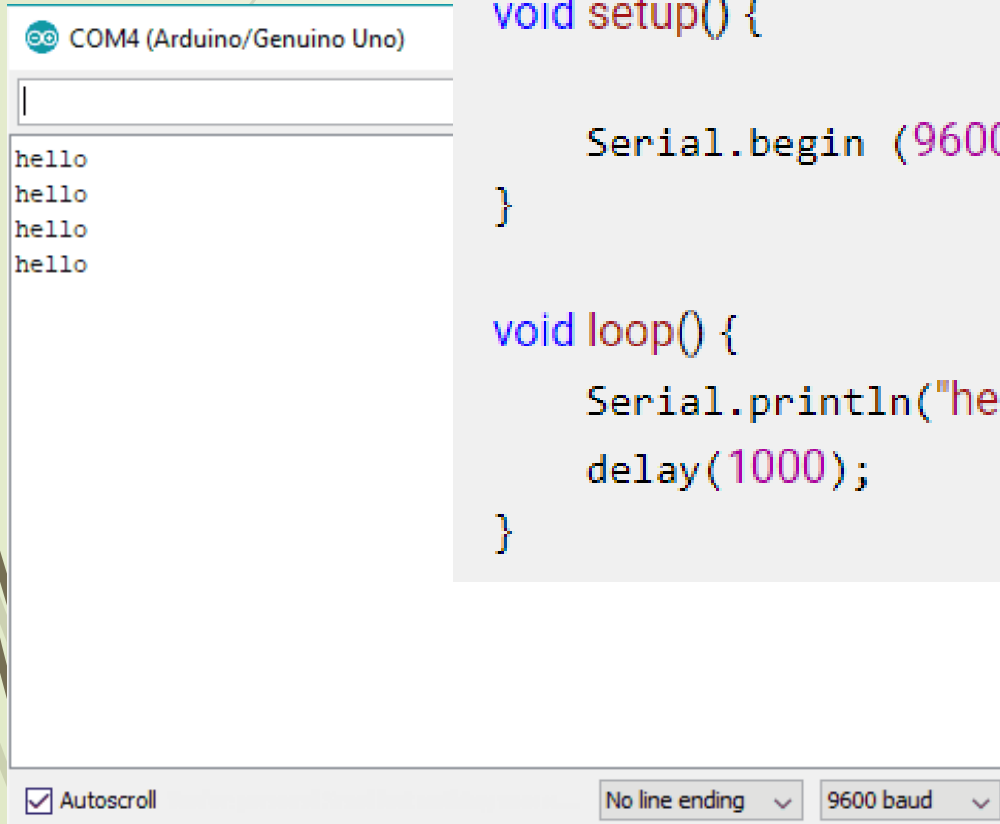
➤ **Serial.read()**

- This function is used to read data serially.
e.g. received_data = Serial.read()

➤ **Serial.readString()**

- This function is used to read the received string.
e.g. String received_data = Serial.readString()

Transmit Data Serially to Computer



```
void setup() {  
    Serial.begin (9600); //initiate serial communication with 9600 baud rate  
}  
  
void loop() {  
    Serial.println("hello"); //send data to computer  
    delay(1000);  
}
```

Read data serially from PC and display on Serial Monitor

```
char received_data;    //variable to store read data
void setup() {
    Serial.begin(9600);
}

void loop() {
    if(Serial.available()>0) //check for any data received
    {
        received_data = Serial.read(); //read received data
        Serial.print("received data is: ");
        Serial.println(received_data); //display received data
    }
}
```

Read transmitted String from PC

```
String receive_buffer;
void setup() {
  Serial.begin(9600);
}

void loop() {
  if(Serial.available()>0) //check for any data received
  {
    receive_buffer = Serial.readString(); //read received data
    Serial.print("received data is: ");
    Serial.println(receive_buffer);      //display received data
  }
}
```

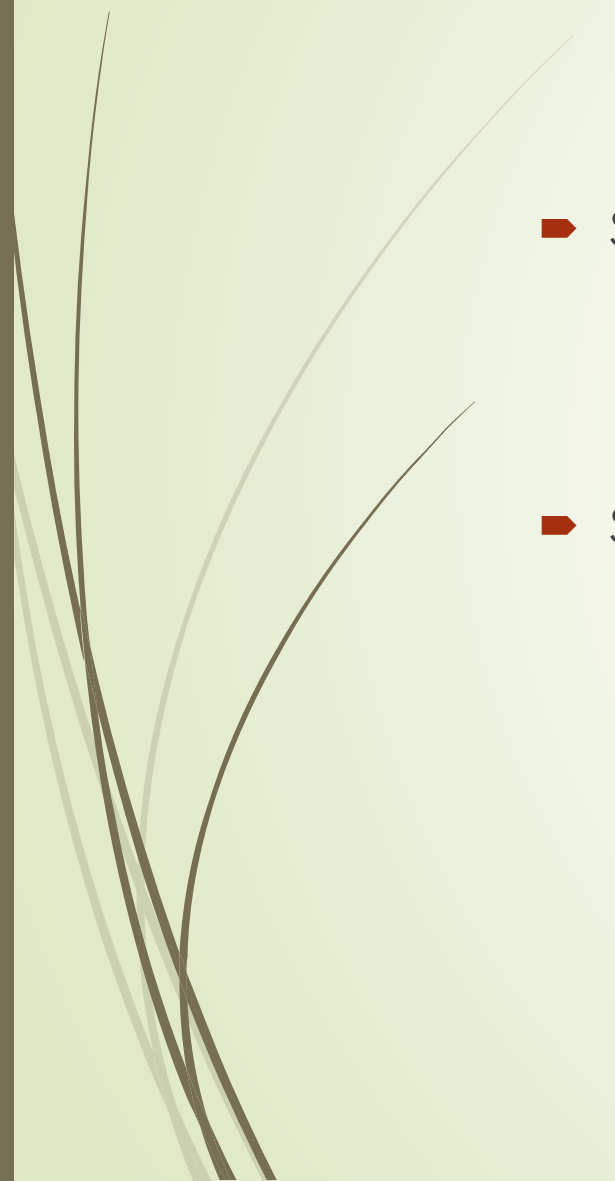
Control LED ON-OFF from PC using Arduino

```
char received_data;  
void setup() {  
    Serial.begin(9600);  
    pinMode(13, OUTPUT);  
}
```

```
void loop() {  
    if(Serial.available()>0){  
        //read serial data  
        received_data = (char)Serial.read();  
  
        Serial.print("received data is:");  
  
        //check for received command for action  
        if(received_data=='1'){  
            digitalWrite(13, HIGH);  
            Serial.println(received_data);  
        }  
        else if(received_data=='0'){  
            digitalWrite(13, LOW);  
            Serial.println(received_data);  
        }  
        else  
            Serial.println(" invalid");  
    }  
}
```

Number Counting

```
void setup() {  
  // put your setup code here, to run once:  
  Serial.begin(9600);  
}  
int i = 0;  
void loop() {  
  // put your main code here, to run repeatedly:  
  Serial.print( "i = " ); // //在監控視窗中顯示訊息  
  Serial.println(i); //在監控視窗中顯示訊息，並換行  
  i++;  
  delay(500);  
}
```



➤ Serial.print()

➤ <https://www.arduino.cc/reference/en/language/functions/communication/serial/print/>

➤ Serial.println()

➤ <https://www.arduino.cc/reference/en/language/functions/communication/serial/println/>